

$$1. \quad B = \{ v_1 = (1, 1, 1), v_2 = (1, 1, -1), v_3 = (1, -1, -1) \}$$

$$e_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{3}} (1, 1, 1)$$

$$e_2' = v_2 - \langle v_2, e_1 \rangle e_1$$

$$= (1, 1, -1) - \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} (1, 1, 1)$$

$$= (1, 1, -1) - \frac{1}{3} (1, 1, 1) = \frac{2}{3} (1, 1, -2) \Rightarrow \|e_2'\| = \frac{2}{3} \sqrt{6}$$

$$e_2 = \frac{e_2'}{\|e_2'\|} = \frac{1}{\sqrt{6}} (1, 1, -2)$$

$$e_3' = v_3 - \langle v_3, e_1 \rangle e_1 - \langle v_3, e_2 \rangle e_2$$

$$= (1, -1, -1) + \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} (1, 1, 1) - 2 \cdot \frac{1}{\sqrt{6}} \cdot \frac{1}{\sqrt{6}} (1, 1, -2)$$

$$= (1, -1, -1) + \frac{1}{3} (1, 1, 1) - \frac{2}{3} (1, 1, -2) = (1, -1, 0) \Rightarrow \|e_3'\| = \sqrt{2}$$

$$e_3 = \frac{e_3'}{\|e_3'\|} = \frac{1}{\sqrt{2}} (1, -1, 0)$$

$$\text{Adăugăm } \{ e_1, e_2, e_3 \} \in \mathbb{R}^3$$

baza ortonormată obținută prin p.o.g.s. din baza dată

$$2. \quad B' = \{ e_1 = \frac{1}{\sqrt{2}} (0, 1, 1), e_2 = \frac{1}{\sqrt{6}} (2, -1, 1), e_3 = \frac{1}{\sqrt{3}} (1, 1, -1) \}$$

$$w = (-1, 1, 2)$$

scriem vectorul w în baza B' dată:

$$w = w_1 e_1 + w_2 e_2 + w_3 e_3$$

$$\langle w, e_1 \rangle = w_1 \underbrace{\langle e_1, e_1 \rangle}_{=1} + w_2 \underbrace{\langle e_2, e_1 \rangle}_{=0} + w_3 \underbrace{\langle e_3, e_1 \rangle}_{=0}$$

$$\Rightarrow w_1 = \langle w, e_1 \rangle = \frac{3}{\sqrt{2}}$$

$$\langle w, e_2 \rangle = w_1 \underbrace{\langle e_1, e_2 \rangle}_{=0} + w_2 \underbrace{\langle e_2, e_2 \rangle}_{=1} + w_3 \underbrace{\langle e_3, e_2 \rangle}_{=0}$$

$$\Rightarrow w_2 = \langle w, e_2 \rangle = \frac{(-2 - 1 + 2)}{\sqrt{6}} = -\frac{1}{\sqrt{6}}$$

$$\langle w, e_3 \rangle = w_1 \underbrace{\langle e_1, e_3 \rangle}_{=0} + w_2 \underbrace{\langle e_2, e_3 \rangle}_{=0} + w_3 \underbrace{\langle e_3, e_3 \rangle}_{=1}$$

$$\Rightarrow w_3 = \langle w, e_3 \rangle = \frac{(-1 + 1 - 2)}{\sqrt{3}} = -\frac{2}{\sqrt{3}}$$

$$3. \quad B = \{ v_1 = (2, 2, 1), v_2 = (-2, -1, 2), v_3 = (5, -6, 2) \}$$

$$e_1 = \frac{v_1}{\|v_1\|} = \frac{1}{3} (2, 2, 1)$$

$$e'_2 = v_2 - \langle v_2, e_1 \rangle e_1$$

$$= (-2, -1, 2) - \frac{(-4)}{3} \cdot \frac{1}{3} (2, 2, 1)$$

$$= \left(-\frac{10}{3}, -\frac{1}{3}, \frac{22}{3} \right) = \frac{1}{3} (-10, -1, 22)$$

$$\|e'_2\| = \frac{1}{3} \cdot \sqrt{100 + 1 + 484} = \frac{1}{3} \cdot \sqrt{585} = \frac{1}{3} \cdot \sqrt{65}$$

$$e_2 = \frac{e'_2}{\|e'_2\|} = \frac{1}{\sqrt{65}} (-10, -1, 22)$$

$$e'_3 = v_3 - \langle v_3, e_1 \rangle e_1 - \langle v_3, e_2 \rangle e_2$$

$$= (5, -6, 2) - 0 \cdot e_1 - 0 \cdot e_2$$

$$= (5, -6, 2)$$

$$\|e'_3\| = \sqrt{25 + 36 + 4} = \sqrt{65}$$

$$e_3 = \frac{e'_3}{\|e'_3\|} = \frac{1}{\sqrt{65}} (5, -6, 2)$$

Având $\{e_1, e_2, e_3\} \subset \mathbb{R}^3$

bază ortonormală sct, prin p.p.s. din baza dată.

$$4. a) U = \langle \underbrace{(1, 2, 1)}_{u_1}, \underbrace{(1, -1, 2)}_{u_2} \rangle$$

$$U^\perp = \{ y \in \mathbb{R}^3 / y \perp x, (\forall) x \in U \}$$

$$B = \{ u_1, u_2 \} \subset U$$

bază

$$\text{Scrie } y = (y_1, y_2, y_3) \in \mathbb{R}^3 \text{ a } \hat{=} \begin{cases} y \perp u_1 \\ y \perp u_2 \end{cases} \Leftrightarrow \begin{cases} \langle y, u_1 \rangle = 0 \\ \langle y, u_2 \rangle = 0 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} y_1 + 2y_2 + y_3 = 0 \\ y_1 - y_2 + 2y_3 = 0 \end{cases}$$

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 2 \end{pmatrix}$$

$$\Delta = \begin{vmatrix} 1 & 2 \\ 1 & -1 \end{vmatrix} = -3 \neq 0 \Rightarrow \text{rang } A = 2$$

$$\begin{cases} y_1, y_2 \text{ nec nec} \\ y_3 = \alpha \text{ nec nec} \end{cases} \Rightarrow \begin{cases} y_1 + 2y_2 = -\alpha \\ y_1 - y_2 = -2\alpha \end{cases} \Rightarrow \begin{cases} y_2 = \frac{\alpha}{3} \\ y_1 = -\frac{5\alpha}{3} \end{cases}$$

$$\text{deci } y = \alpha \left(-\frac{5}{3}, \frac{1}{3}, 1 \right), \alpha \in \mathbb{R}$$

$$U^\perp = \left\{ \alpha \left(-\frac{5}{3}, \frac{1}{3}, 1 \right) / \alpha \in \mathbb{R} \right\} = \left\langle \left(-\frac{5}{3}, \frac{1}{3}, 1 \right) \right\rangle$$

$\Rightarrow$ complementul ortogonal al lui U (sp. vect 1-dim)

$$b) V = \langle \underbrace{(2, -3, 1)}_v \rangle$$

$$V^\perp = \{ y \in \mathbb{R}^3 / y \perp x, (\forall) x \in V \}$$

$$B = \{ v \} \subset V$$

bază

fie $y = (y_1, y_2, y_3) \in \mathbb{R}^3$ a $\hat{c} \quad y \perp v \Leftrightarrow \langle y, v \rangle = 0 \Leftrightarrow$

$$\Leftrightarrow 2y_1 - 3y_2 + y_3 = 0$$

$$\text{deci } v^\perp = \{ y \in \mathbb{R}^3 \mid 2y_1 - 3y_2 + y_3 = 0 \}$$

(y_1, y_2, y_3)

→ mulțime ortogonală la lui v (spațiu vectorial 2-dimensional)